Ethics, Fairness, Accountability, and Transparency in ML

Selpi

Applied Machine Learning

Acknowledgement

- Sumpter (2021) Ethics in machine learning, In: *Machine Learning: A First Course for Engineers and Scientists*, Cambridge University Press
- Google course on fairness in ML
- Introduction to interpretability
- Interpretability
- Richard Johansson’s earlier slides

Why care about ethics and FAT in ML?

FAT (Fairness, Accountability, and Transparency)

- Because we all have bias and are bias
- Implications
- People's rights and obligations

Ethics, FAT in ML

What comes to mind?

- Privacy:
- Bias
- Consent
- Unwanted effect
- Trust

Discussion

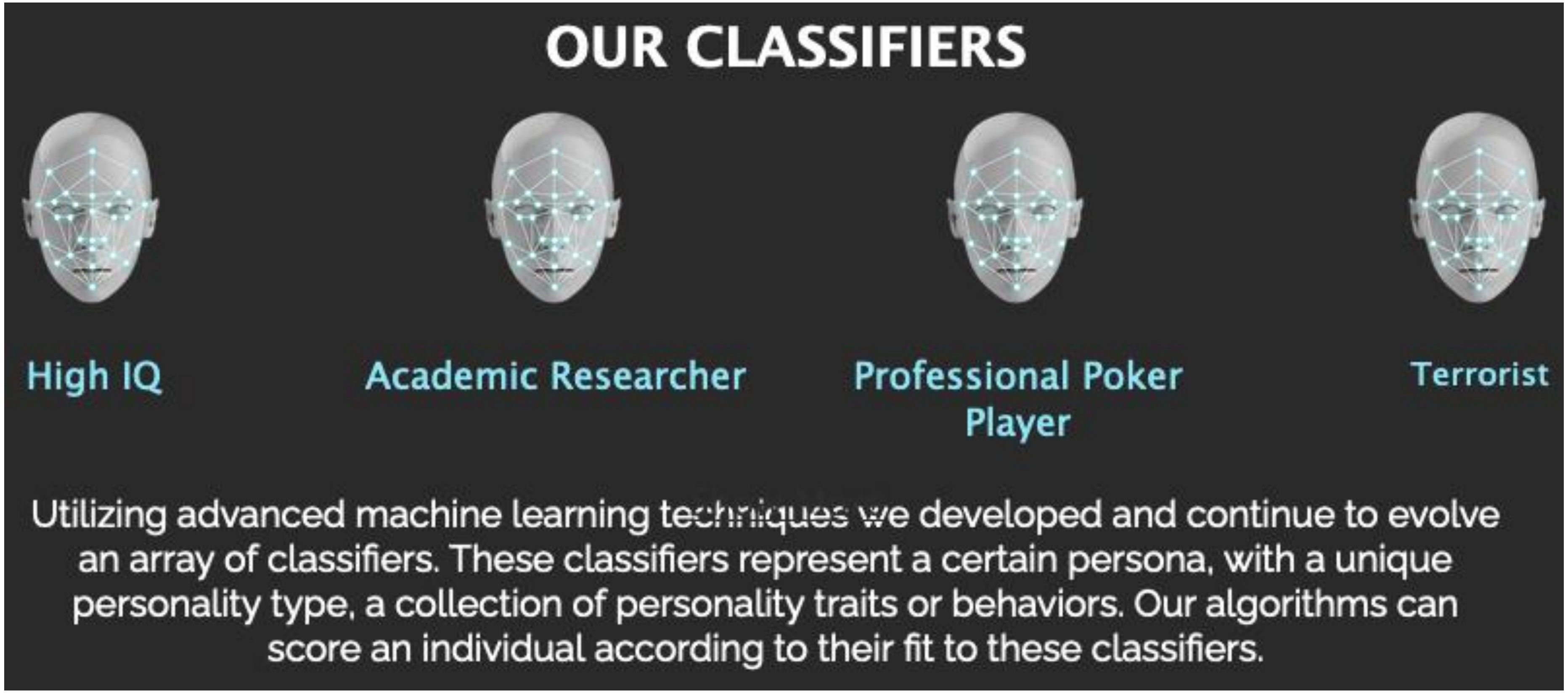
For each of the following cases:

- What are the issues (related to ethical, fairness, accountability, and transparancy) that have to be considered?
- What are the potential implications of having such a system?
- As someone who knows how an ML system is built in general, are there relevant technical questions that you would ask related to how the ML system has been built?

Controversial software claims to tell personality from your face

A start-up says its face-recognition tech can identify people's personality type from photos – and spot terrorists, paedophiles and poker players in crowds

Source



?

Source

More cases

- Pizza checker

- ChatGPT

- Use of AI for war related

Types of bias

What might be the issues in the following cases?

- Measure product quality based on reviews sent in by customers
- Train a model to predict future sales of a new product based on phone surveys on current customers
- Train a model using data from the first x customers who responded to an email

Types of bias

- Reporting bias
  - Only send in review if we really like or really don't like something
- Automation bias
- Selection bias
  - Coverage bias
  - Non-response bias
  - Sampling bias
- Group attribution bias
  - In group bias
  - Out-group homogeneity bias
- Implicit bias
  - Confirmation/Experimenter's bias
- Many others…

Evaluating fairness

- Your task: evaluate the fairness of an ML model (is the model fair?)
- The model:
  - was built to find people who might be interested to study at a university in Sweden, based on their activity on a social networking site.
  - recommends or does not recommend the university to users.
  - was tested on 600 non-Swedes and 1200 Swedes (all 1800 have given permission to use their data, and would be eligible to go to university)

Non-Swedes	Not Interested ($y = -1$)	Interested ($y = 1$)
Not recommended course ($\widehat{y}(\mathbf{x}) = -1$)	TN = 300	FN = 100
Recommended course ($\widehat{y}(\mathbf{x}) = 1$)	FP = 100	TP = 100
Swedes	Not Interested ($y = -1$)	Interested ($y = 1$)
Not recommended course ($\widehat{y}(\mathbf{x}) = -1$)	TN = 400	FN = 50
Recommended course ($\widehat{y}(\mathbf{x}) = 1$)	FP = 350	TP = 400

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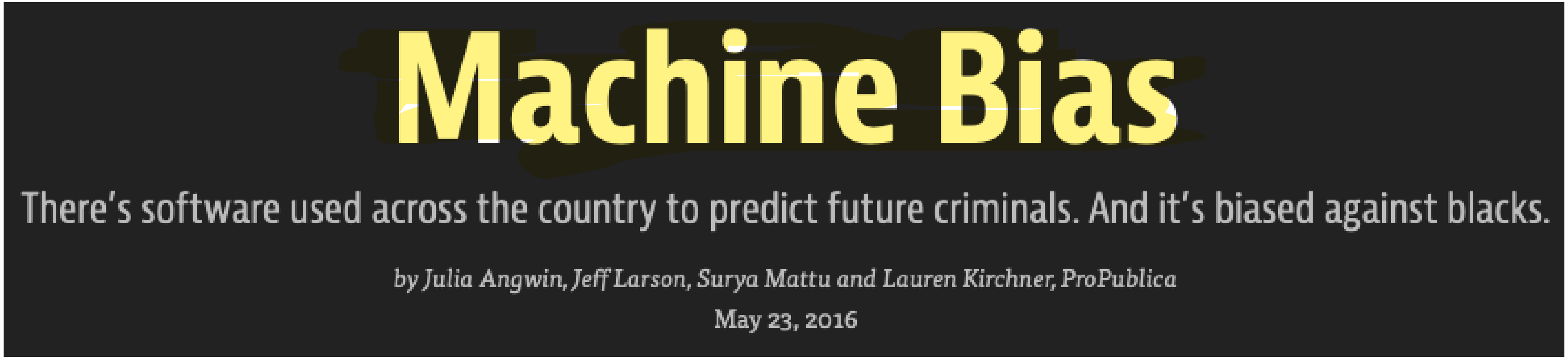
- Misclassification error = (100 + 100)/600 = 1/3 for non-Swedes and (50+350)/1200 = 1/3 for Swedes.
- False negative rate (miss rate) = = FN/(TP/FN) => 100/(100+100) = ½ for non-Swedes and 50/(400+50) =1/9 for Swedes.
- False positive rate = FP/(TN+FP) => 300/(300+100) = 1/4 for non-Swedes and 350/(400+350)=7/15 for Swedes

- Is the model fair?



Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica

May 23, 2016

Compas algorithm by Northpointe

Source

- To help with criminal sentencing decision
- Discussion on for or against the algorithm (see here)
- Model used logistic regression with more than 100 input variables;
  - age at first arrest, years of education, and questionnaire answers about family background, drug use and other factors to predict, an output variable, as to whether the person would reoffend
  - race was not included
- Is the model fair? How to evaluate this?

Compas algorithm by Northpointe

Black defendants	Didn't reoffend ($y = -1$)	Reoffended ($y = 1$)
Lower risk ($\widehat{y}(\mathbf{x}) = -1$)	TN = 990	FN = 532
Higher risk ($\widehat{y}(\mathbf{x}) = 1$)	FP = 805	TP = 1369
White defendants	Didn't reoffend ($y = -1$)	Reoffended ($y = 1$)
Lower risk ($\widehat{y}(\mathbf{x}) = -1$)	TN = 1139	FN = 461
Higher risk ($\widehat{y}(\mathbf{x}) = 1$)	FP = 349	TP = 505

- False positive rate= FP/(TN+FP) => 805/(990+805) = 44.8% for black defendants and 349/(1139+349)=23.4% for white defendants
- True positive rate = TP/(FN+TP) => 1369/(532+1369) = 72% for black defendants and 505/(461+505) = 52.2% for white defendants.
- Precision?

Summary - Fairness

- "No single function for measuring fairness" (Sumpter 2021)
- "Complete fairness is mathematically impossible" (Sumpter 2021)
- Fairness through awareness (Dwork et al. 2012)

Other sources of fairness/ethical issues in ML

- Training data
  - Word embeddings
- Funding related
- Need to publish
- Competing companies/individuals

Interpretability - Definitions

- "Interpretability is the degree to which a human can understand the cause of a decision" Miller (2017).
- "Interpretability is the degree to which a human can consistently predict the model's result" Kim et al. (2016).

Interpretability

- Not needed when:
  - Used in a low-risk environment (e.g., recommender system)
  - If the method has been studied and evaluated thoroughly (e.g., character recognition)
- Vs explanationability
- Interpretability in the ML pipeline

Depending on the problem, a high AUC value (e.g., 0.8) is not always good. Discuss why that is the case, with each of the following scenarios:

- ”An algorithm is shown two input images, one containing a cancerous tumour, one not containing a cancerous tumour. The two images are selected at random from those of people referred by a specialist for a scan. AUC is the proportion of times the algorithm correctly identifies the image containing the tumour” (SML book, Ch.12).

- ”An algorithm is shown input data of the location of two randomly chosen shots from a season of football (soccer) and predicts whether the shot is a goal or not. AUC is the proportion of times it correctly identifies the goal.” (SML book, Ch.12)

Privacy preserving ML

- https://ppml-workshop.github.io/

- To allow different organisations to collaboratively train ML models without releasing their private data in its original form

Other issues

- Reliability
- Privacy
- Representativity
- Treatment of workers (in annotation)
- Effects of ML on society
- Responsibility
- Interpretability
- Explainability